The Distribution of Geodashing Dashpoints

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XXXXXXXX Geographic Information Analysis

University of XXXXXXX

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Introduction

The number and popularity of GPS games has increased significantly over the past several years. With the increasing capability of smart phones and rising popularity of social media and location based services, casual users who previously would not have had an interest in location based activities or games are now easily and intuitively engaged through normal interactions with their smart devices.

One of these GPS games is Geodashing which will celebrate its 10th anniversary in June 2011. Geodashing is a game in which players use GPS receivers on a playing field that covers the entire planet. The waypoints, or dashpoints, to be reached are randomly selected. The win goes to the individual or team that can get to the most dashpoints; that is, if you can get to them at all! Each month, a new set of dashpoints is created making each game completely different and unpredictable. (Scout 2011)

"Getting there is all the fun" is the official slogan and that is what makes it challenging. Scoring involves getting to within 100 meters of WGS-84 coordinates and then describing it in enough detail to prove the visit. Entertaining stories and pictures are encouraged. Some dashpoints are very remote and completely unscorable, while others are classified as drive-bys where you can simply drive through the point. A favorable distribution of dashpoints can lead to a high score, conversely an unfavorable distribution can lead to very long journeys for a single scorable point. There are 739 players worldwide. Among the active players, a recurring topic is the monthly dashpoint distribution. Are these distributions really random; why are some months so much better than others; and why do some players always seem to get dashpoints located close to them?

Various geographic information analytical tools will be used to determine if the monthly distribution is really random or are they clustered. The research question will be

stated as dashpoints exhibit spatial patterns and clustering. The null hypothesis will be stated as dashpoints do not exhibit spatial patterns and clustering.

Background

Several interesting journal articles were found describing the use of GPS tracking to analyze the paths of tourists (Pettersson and Zillinger 2011), or family members (van der Spek, et al. 2009) to describe human activity. Use of recreational GPS data included determining which positions on soccer teams had the most activity (Gray, et al. 2010), and some incredible artwork created by walking with a GPS (Lauriault 2009). All described the potential uses of GPS position track data and how it may be collected automatically and possibly without the knowledge of the user on their smart devices. No articles were found analyzing the growth and social impact of GPS games.

The creator of Geodashing describes the monthly dashpoint creation process:

> Starting in Alaska at some predefined waypoint, random offsets are generated to randomize a dashpoint location. Then a waypoint is projected due east at some predefined distance, measured in kilometers, not in longitude. There, more random offsets are generated to randomize a second dashpoint location. This stepping east and randomize operation is repeated over and over until we're back in Alaska where we started. Then, another waypoint is projected due south and start the stepping east and randomize operation all over again. The number of steps around the globe increases with decreasing latitude. The longitude of the steps don't line up exactly with the steps farther north, because the Earth's circumference grows as you move from the pole to the Equator. Each time a random dashpoint is generated, its location is compared with its neighbors. If it's closer than some predefined minimum, then the random dashpoint is discarded and a new candidate is generated. This is repeated until either a suitable dashpoint is found or some arbitrary number of attempts is reached. The process stops south of New Zealand. All the generated dashpoints are reviewed for whether they ended up on land or on water. The water dashpoints are discarded. The ones that are left are ordered west to east and given alphabetical names from a predefined dictionary (Scout 2011).

Will spatial analysis uncover clustering or find complete spatial randomness in the game data?

Methods

The state of Arizona was chosen as the study area. 119 zip files, each containing approximately 32,000 latitude/longitude pairs of unprojected WGS-84 data in CSV format were downloaded. A macro was developed and refined in Microsoft Excel to parse each games data to an area slightly larger than the state of Arizona and then save the file of approximately 100 points to a *.xlsx format for import into ArcGIS.

Game data would be combined and processed as three datasets for analysis: a single month; a three year period; and the totality of points for the life of the game. A sample size of at least 30 is generally recognized as the minimum appropriate for analysis. A three year period (2007-2010) was selected for this analysis to contrast against data from a single month (May 2011). All game data would then be used to a further confirm results. Since each point has no value other than its location attribute, some Geostatistical Anayst Explore Data tools such as the Spatial Autocorrelation (Morans I) Tool were deemed inappropriate for analysis.

Prior to processing, the Project Tool was used to convert the three datasets into a NAD_1983_UTM_Zone_12N projection for analysis. A 20 km buffer was placed around the state of Arizona and this outline was used to Clip all datasets and was also used as a Study Area Feature for the Ripleys K Function Tool.

The Directional Distribution (Standard Deviation Ellipse) Tool, Create Thiessen Polygons Tool, Multi Distance Spatial Cluster Analysis (Ripleys K Function) Tool, Average Nearest Neighbor Tool, and Point Density Analysis Tools would be used for analysis.

Results

Figure 1 depicts the May2011 dashpoint distribution and associated Thiessen Polygons. It also depicts the extents of data used for all processing, with the Arizona border in green and the buffer in black. This distribution is typical for all individual games. The Directional Distribution ellipses from all datasets showed only minor deviations from each other.

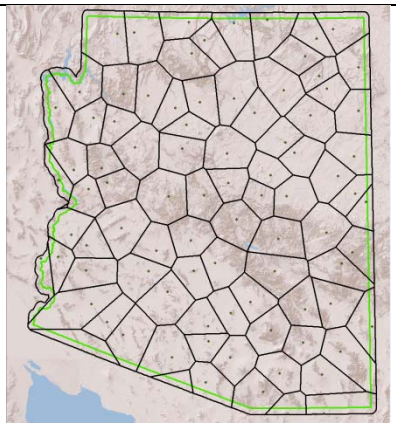


Figure 1 May 2011 Thiessen Polygons

Figure 2 shows the distribution of the raw datasets. There are approximately 100 points per game that fall within the bounded area.

Figure 3 shows the output of Ripleys K Function. 10 Distance Bands and 9 Permutations were used for analysis. The May 2011 data shows statistically significant dispersion at distances less than 67 km. The 3 Years and All Games data show dispersion over the full range of distances that is not statistically significant. This data was plotted in Excel from the ArcGIS results when the Display Results Graphically check box failed to produce an output graph.

Figure 4 shows the Average Nearest Neighbor Summaries. The May 2011 data shows a dispersed pattern with a z-score of 5.1, the 3 Years and All Games results show clustered data with z-scores of -6.3 and -12.3.

Figure 5 shows the Point Density Analysis with results much like the Average Nearest Neighbor: dispersed data for a single month but, clustered data over time.

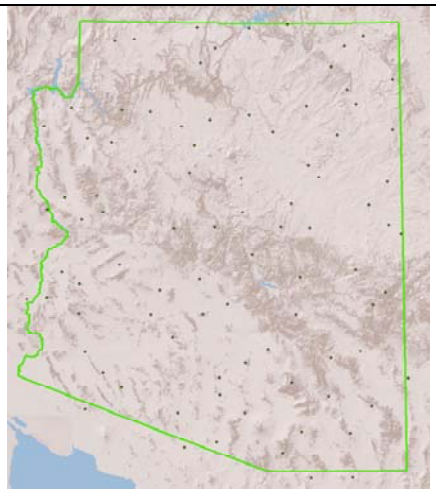
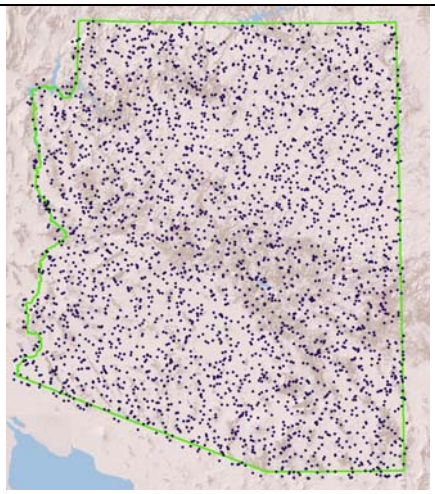
May 2011	3 Years	All Games
		

Figure 2 Distribution of Dashpoints

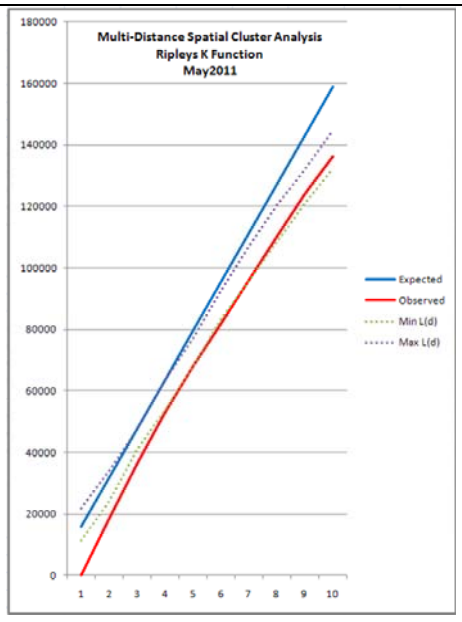
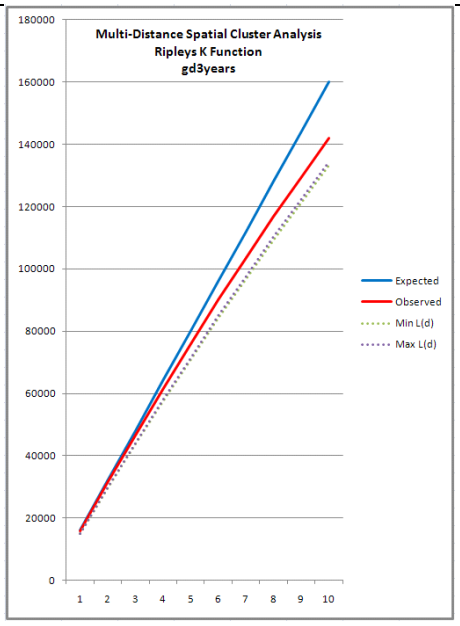
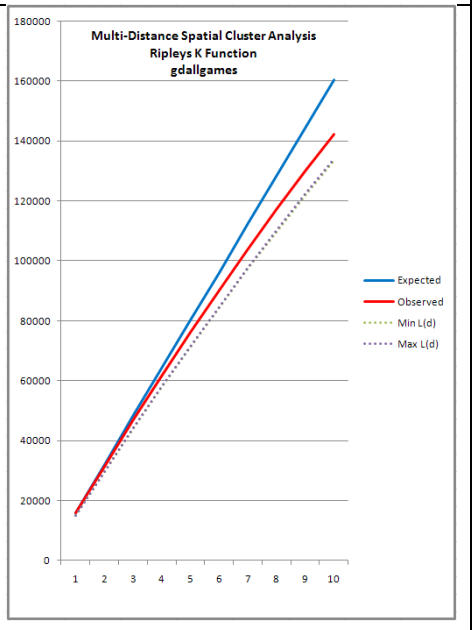
May 2011	3 Years	All Games
		

Figure 3 Ripleys K Function

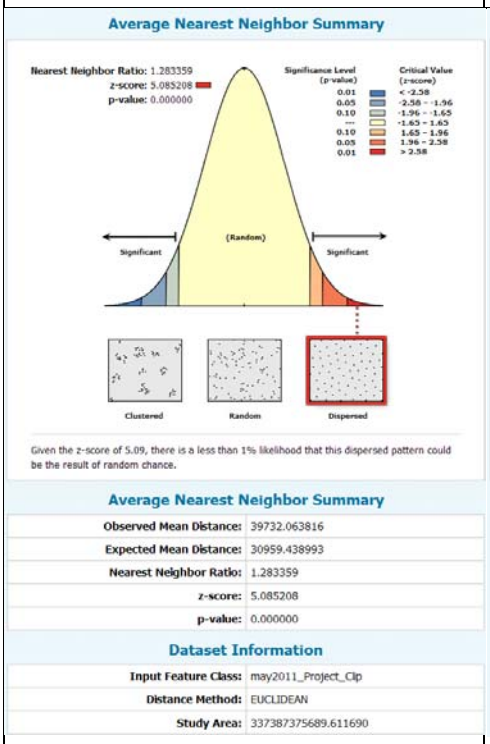
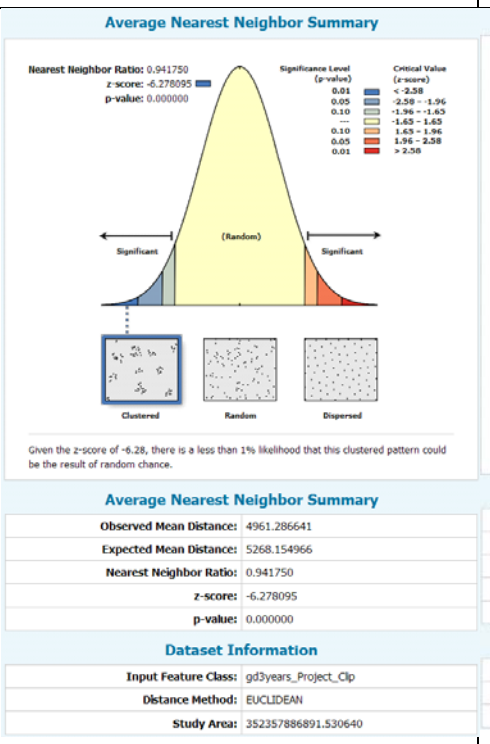
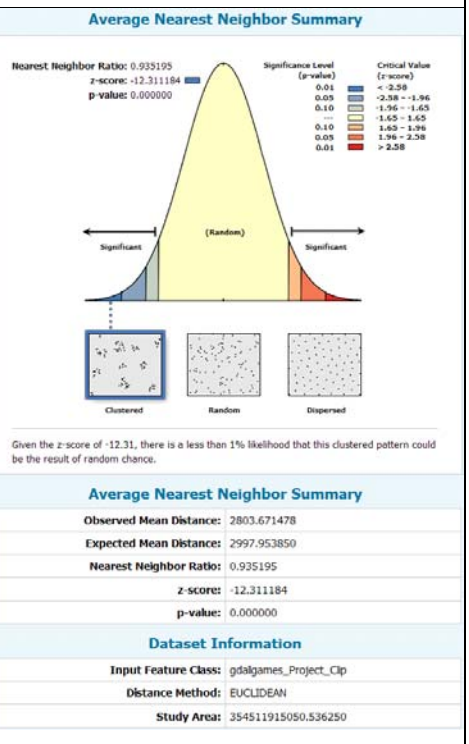
May 2011	3 Years	All Games
		
Average Nearest Neighbor Summary	**Average Nearest Neighbor Summary**	**Average Nearest Neighbor Summary**
Observed Mean Distance: 39732.063816 Expected Mean Distance: 30959.438993 Nearest Neighbor Ratio: 1.283359 z-score: 5.085208 p-value: 0.000000	Observed Mean Distance: 4961.286641 Expected Mean Distance: 5268.154966 Nearest Neighbor Ratio: 0.941750 z-score: -6.278095 p-value: 0.000000	Observed Mean Distance: 2803.671478 Expected Mean Distance: 2997.953850 Nearest Neighbor Ratio: 0.935195 z-score: -12.311184 p-value: 0.000000
Dataset Information	**Dataset Information**	**Dataset Information**
Input Feature Class: may2011_Project_Clip Distance Method: EUCLIDEAN Study Area: 337387375689.611690	Input Feature Class: gd3years_Project_Clip Distance Method: EUCLIDEAN Study Area: 352357886891.530640	Input Feature Class: gdallgames_Project_Clip Distance Method: EUCLIDEAN Study Area: 354511915050.536250

Figure 4 Average Nearest Neighbor

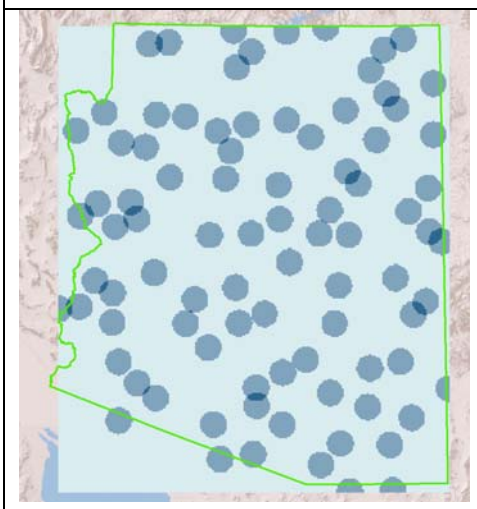
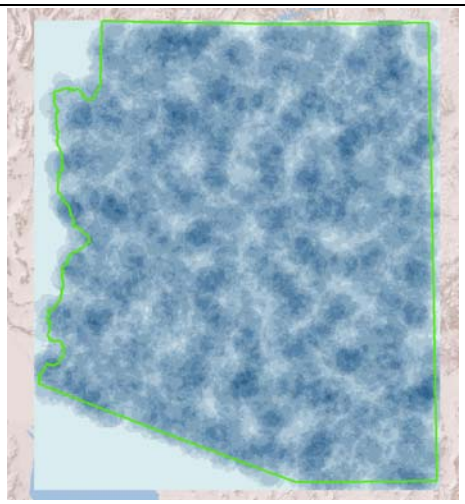
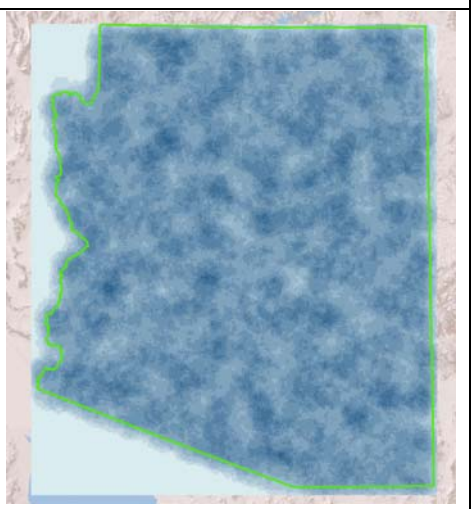
May 2011	3 Years	All Games
		

Figure 5 Point Density Analysis

Discussion

Ripleys K Function summarizes clustering or dispersion over a range of distances. Statistically significant dispersion was noted in the May 2011 dataset. The other 2 datasets did not show significant dispersion or clustering but tended towards dispersion.

The Average Nearest Neighbor computed a z-score of 5.085208 for the May2011 dataset shows almost completely dispersed data with less than a 1% likelihood that the dispersed pattern could be the result of random chance. The same tool used on the 3 year data set shows almost completely the opposite. With a z-score of -6.278095, there is less than a 1% likelihood that this clustered pattern could be the result of random chance. The total population of dashpoints produces a z-score of -12.311184 with less than 1% likelihood that this clustered pattern could be the result of random chance. In other words, with a standard deviation score of -12.311184, the confidence level is 99% that this reflects clustered data. Spatial clustering increases with the number of games. This was depicted graphically using the Point Density Analysis Tool in Figure 5.

Conclusion

The null hypothesis has been proven false. While the dashpoints for individual games are strongly dispersed, geographic clustering increases over time. Without considering other geographic features that may prevent reaching dashpoints, such clustering may be a strong advantage to players located in the higher density zones and a distinct disadvantage to players located in the lower density zones. This may explain why a single month appears to have completely dispersed points, yet month-to-month many dashpoints are located in the same general area, often prompting comments from players about the proximity to previously scored dashpoints.

List of References

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van der Spek, Stefan, Jeroen van Schaick, Peter de Bois, and Remco de Haan. "Sensing Human Activity: GPS Tracking." *Sensors* 9 (2009): 3033-3055.